

Target validation paths — cca-mets-scope-unknown-z3p9

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

This case hinges on one test that has not been run: comprehensive genomic profiling. No molecular data was supplied, so every targeted option is unmeasured and a single paired DNA plus RNA panel is the gate for all of them. If that panel and the two protein assays beside it all return negative, this report has no within-scope recommendations, and the next conversation about standard care for metastatic cholangiocarcinoma belongs to the treating team rather than to Libby.

Comprehensive genomic profiling

The essential first move is one paired DNA plus RNA panel with structural-variant and fusion calling, MSI, and TMB. It resolves seven of the eight candidate axes at once: FGFR2 fusion, IDH1 R132, BRAF V600E, KRAS G12C, NTRK1/2/3 and RET fusion, MSI and TMB. This panel gates pemigatinib, futibatinib, ivosidenib, dabrafenib plus trametinib, selpercatinib, larotrectinib, entrectinib, and pembrolizumab. The RNA arm earns its place because DNA-only calling misses a subset of FGFR2 and NTRK/RET fusions with intronic breakpoints, and the IDH1 R132 and KRAS codon-12 hotspots can fall outside a fusion-focused panel, so confirm the order covers both DNA hotspots and an RNA fusion read. Tissue turnaround is 2 to 3 weeks; ctDNA returns in 7 to 10 days as a faster parallel option when tissue is limited, at the cost of lower fusion sensitivity. Archival FFPE is the default and a fresh biopsy is not required if stored tissue is adequate.

HER2 (ERBB2)

HER2 keys off a protein read, not the DNA panel. The essential assay is HER2 IHC on the PATHWAY 4B5 clone with reflex ISH. An NGS copy-number call does not establish protein expression, and the approved biliary HER2 agents are written against the IHC/ISH result: zanidatamab via NCT04466891 was defined by ISH amplification plus IHC 2+/3+, and trastuzumab deruxtecan is labeled for IHC 3+. The 4B5 antibody is the companion diagnostic, and this assay decides which anti-HER2 agent applies. Turnaround is 3 to 7 days.

MSI-H / dMMR

Mismatch-repair IHC across MLH1, MSH2, MSH6, and PMS2 with reflex MSI by PCR is the high-priority orthogonal read that gates pembrolizumab. It is the fastest and cheapest way to confirm dMMR and catches low-purity cases an NGS MSI score can miss. Concordance with the NGS read is high but not perfect, so the two together harden the call before committing to checkpoint blockade. A clean negative redirects attention to the targeted axes. IHC turnaround is 3 to 5 days, with reflex MSI-PCR about a week.

BRAF V600E

This is captured by the comprehensive panel, so it usually needs no separate order. The high-priority caveat is that a VE1 IHC screen alone is suggestive only: the dabrafenib plus trametinib label and the ROAR cohort were built on a molecularly confirmed V600E, so a positive IHC needs NGS or PCR confirmation, and a non-V600E class-2/3 BRAF alteration does not qualify for the same regimen.

NTRK / RET, KRAS G12C, and downstream workup

The RNA arm of the comprehensive panel also resolves the rare NTRK and RET fusions that gate larotrectinib, entrectinib, and selpercatinib, and the DNA arm reports KRAS G12C. Lower-priority items round out the plan: serial ctDNA tracks acquired FGFR2 kinase-domain resistance once an FGFR inhibitor is running, and a germline referral follows only when MMR IHC shows MLH1 loss or the somatic panel flags a Lynch or BRCA finding, which carries family implications without changing immediate therapy.

Where to order these assays

Assay	Provider	Decision gated	Contact
Comprehensive genomic profiling: paired DNA + RNA NGS with structural-variant/fusion calling, MSI, and TMB	Foundation Medicine (preferred) (FoundationOne CDx (tissue) / FoundationOne Liquid CDx)	Gates all targeted second-line therapy in metastatic CCA (FGFR, IDH1, HER2, BRAF, RET, NTRK inhibitors and pembrolizumab); shared first-step workup.	test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639
Comprehensive genomic profiling: paired DNA + RNA NGS with structural-variant/fusion calling, MSI, and TMB	Tempus (Tempus xT (DNA) + xR (RNA fusion))	Gates all targeted second-line therapy in metastatic CCA (FGFR, IDH1, HER2, BRAF, RET, NTRK inhibitors and pembrolizumab); shared first-step workup.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
Comprehensive genomic profiling: paired DNA + RNA NGS with structural-variant/fusion calling, MSI, and TMB	Caris Life Sciences (Caris MI Profile (whole-exome + whole-transcriptome))	Gates all targeted second-line therapy in metastatic CCA (FGFR, IDH1, HER2, BRAF, RET, NTRK inhibitors and pembrolizumab); shared first-step workup.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 888-979-8669
Comprehensive genomic profiling: paired DNA + RNA NGS with structural-variant/fusion calling, MSI, and TMB	Guardant Health (Guardant360 CDx (ctDNA))	Gates all targeted second-line therapy in metastatic CCA (FGFR, IDH1, HER2, BRAF, RET, NTRK inhibitors and pembrolizumab); shared first-step workup.	test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887
HER2 IHC (PATHWAY anti-HER2/neu clone 4B5) with reflex HER2 ISH	NeoGenomics Laboratories (preferred) (HER2/neu (4B5) IHC with reflex ISH)	Zanidatamab (HERIZON-BTC-01) and trastuzumab deruxtecan (IHC 3+); selects which anti-HER2 agent applies.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907
HER2 IHC (PATHWAY anti-HER2/neu clone 4B5) with reflex HER2 ISH	Labcorp (HER2 (HER-2/neu) IHC and FISH)	Zanidatamab (HERIZON-BTC-01) and trastuzumab deruxtecan (IHC 3+); selects which anti-HER2 agent applies.	test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167
HER2 IHC (PATHWAY anti-HER2/neu clone 4B5) with reflex HER2 ISH	Quest Diagnostics (HER2 IHC with reflex ISH)	Zanidatamab (HERIZON-BTC-01) and trastuzumab deruxtecan (IHC 3+); selects which anti-HER2 agent applies.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 866-697-8378

Mismatch-repair IHC (MLH1, MSH2, MSH6, PMS2) with reflex MSI by PCR	NeoGenomics Laboratories (preferred) (MMR IHC panel with reflex MSI)	Pembrolizumab under the MSI-H/dMMR tumor-agnostic label; orthogonal confirmation of any NGS MSI signal.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907
Mismatch-repair IHC (MLH1, MSH2, MSH6, PMS2) with reflex MSI by PCR	Labcorp (MMR IHC / MSI by PCR)	Pembrolizumab under the MSI-H/dMMR tumor-agnostic label; orthogonal confirmation of any NGS MSI signal.	test info · 358 South Main Street, Burlington, NC 27215 · 800-845-6167
Mismatch-repair IHC (MLH1, MSH2, MSH6, PMS2) with reflex MSI by PCR	Mayo Clinic Laboratories (Mismatch repair (MMR) IHC and MSI)	Pembrolizumab under the MSI-H/dMMR tumor-agnostic label; orthogonal confirmation of any NGS MSI signal.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 800-533-1710
BRAF V600E confirmation by NGS (or VE1 IHC screen confirmed by NGS/PCR)	Foundation Medicine (preferred) (FoundationOne CDx)	Dabrafenib + trametinib (BRAF V600E tumor-agnostic / ROAR); distinguishes V600E from non-actionable class-2/3 BRAF.	test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639
BRAF V600E confirmation by NGS (or VE1 IHC screen confirmed by NGS/PCR)	NeoGenomics Laboratories (BRAF V600E (VE1) IHC)	Dabrafenib + trametinib (BRAF V600E tumor-agnostic / ROAR); distinguishes V600E from non-actionable class-2/3 BRAF.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907
IDH1 R132 hotspot detection by tumor DNA NGS (included in comprehensive panel)	Foundation Medicine (preferred) (FoundationOne CDx)	Ivosidenib (ClarIDHy) for IDH1 R132-mutant CCA; confirms DNA hotspot coverage in the ordered panel.	test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639
IDH1 R132 hotspot detection by tumor DNA NGS (included in comprehensive panel)	Tempus (Tempus xT)	Ivosidenib (ClarIDHy) for IDH1 R132-mutant CCA; confirms DNA hotspot coverage in the ordered panel.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
RNA-based fusion NGS for NTRK1/2/3 and RET (pan-TRK IHC as an optional screen)	Caris Life Sciences (preferred) (Caris MI Profile (whole-transcriptome))	Larotrectinib/entrectinib (NTRK) and selpercatinib (RET) tumor-agnostic labels; confirms RNA fusion coverage.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 888-979-8669
RNA-based fusion NGS for NTRK1/2/3 and RET (pan-TRK IHC as an optional screen)	Tempus (Tempus xR (RNA fusion))	Larotrectinib/entrectinib (NTRK) and selpercatinib (RET) tumor-agnostic labels; confirms RNA fusion coverage.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 800-739-4137
RNA-based fusion NGS for NTRK1/2/3 and RET (pan-TRK IHC as an optional screen)	ArcherDX / Invitae (Archer FusionPlex)	Larotrectinib/entrectinib (NTRK) and selpercatinib (RET) tumor-agnostic labels; confirms RNA fusion coverage.	test info · 1400 16th Street, San Francisco, CA 94103 · 800-436-3037
KRAS G12C detection by tumor DNA NGS (included in comprehensive panel)	Foundation Medicine (preferred) (FoundationOne CDx)	Adagrasib/sotorasib-class KRAS G12C inhibitors and G12C trial enrollment.	test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639
KRAS G12C detection by tumor DNA NGS (included in comprehensive panel)	Guardant Health (Guardant360 CDx (ctDNA))	Adagrasib/sotorasib-class KRAS G12C inhibitors and G12C trial enrollment.	test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887

Baseline and serial ctDNA for FGFR2 kinase-domain mutations and co-altered PIK3CA / CDKN2A/B	Guardant Health (preferred) (Guardant360)	Switch between FGFR inhibitors (e.g. to next-generation agents) on acquired-resistance signal; does not gate enrollment.	test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887
Baseline and serial ctDNA for FGFR2 kinase-domain mutations and co-altered PIK3CA / CDKN2A/B	Foundation Medicine (FoundationOne Liquid CDx)	Switch between FGFR inhibitors (e.g. to next-generation agents) on acquired-resistance signal; does not gate enrollment.	test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639
Germline multigene panel (Lynch genes BRCA1/2; reflex from somatic findings)	Invitae (preferred) (Invitae Multi-Cancer / Lynch Syndrome Panel)	Hereditary-syndrome diagnosis and cascade family screening; does not change current systemic therapy.	test info · 1400 16th Street, San Francisco, CA 94103 · 800-436-3037
Germline multigene panel (Lynch genes BRCA1/2; reflex from somatic findings)	GeneDx (GeneDx hereditary cancer panel)	Hereditary-syndrome diagnosis and cascade family screening; does not change current systemic therapy.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 888-729-1206

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
Comprehensive genomic profiling: paired DNA + RNA NGS with calling, MSI, and TMB (covering FGFR2, IDH1, ERBB2 copy number, BRAF V600E, KRAS G12C, NTRK1/2/3, RET)	No molecular profiling was supplied, so every actionable axis in this case is unmeasured; a single comprehensive DNA+RNA panel resolves seven of the eight candidate features at once and is the gate for all targeted second-line options. NCCN advises CGP at diagnosis of unresectable/metastatic biliary tract cancer, and an RNA-based fusion read matters because DNA-only calling misses a subset of FGFR2 fusions with novel or intronic breakpoints. Skipping it leaves the patient on chemotherapy alone while a roughly 40% chance of an actionable alteration goes undetected.	Foundation Medicine (FoundationOne CDx (tissue) / FoundationOne Liquid CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639	archival FFPE block or 10-20 unstained slides; or 2 x 10 mL Streck tubes whole blood for ctDNA

HER2 IHC (PATHWAY anti-HER2/neu clone 4B5) with reflex HER2 ISH	NGS copy-number alone does not establish HER2 protein expression, and the approved biliary-tract HER2 agents key off a protein/ISH readout, not an amplification call: trastuzumab deruxtecan is labeled for IHC 3+, and zanidatamab eligibility was defined by ISH amplification plus IHC 2+/3+ in HERIZON-BTC-01. The PATHWAY 4B5 antibody is the FDA companion diagnostic for zanidatamab in biliary tract cancer. Without IHC/ISH, an NGS HER2 signal cannot be converted into a drug choice or the correct agent.	NeoGenomics Laboratories (HER2/neu (4B5) IHC with reflex ISH) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907	archival FFPE block or 4-5 unstained slides
Mismatch-repair IHC (MLH1, MSH2, MSH6, PMS2) with reflex MSI by PCR	MMR protein IHC is the fastest, cheapest orthogonal readout for dMMR and catches cases an NGS-derived MSI score can miss when tumor purity is low; concordance with MSI-PCR is high but discordances occur, so the two together harden the call before committing to checkpoint blockade. The VENTANA MMR Rx Dx panel is an FDA companion diagnostic in the dMMR tumor-agnostic setting. MSI-H/dMMR is uncommon in biliary tract cancer (roughly 1-3%), so a confirmed positive is decision-changing and a clean negative redirects attention to the targeted axes.	NeoGenomics Laboratories (MMR IHC panel with reflex MSI) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 866-776-5907	archival FFPE block or 4 unstained slides

BRAF V600E confirmation by NGS (or VE1 IHC screen confirmed by NGS/PCR)	VE1 IHC is a sensitive screen but is not definitive on its own; the dabrafenib/trametinib tumor-agnostic label and the ROAR biliary cohort were built on molecularly confirmed BRAF V600E, so a positive IHC needs NGS or PCR confirmation before committing. The comprehensive panel already covers this codon, so this row mainly flags that a non-V600E BRAF class-2/3 alteration would not qualify for the same regimen. Treating on an unconfirmed IHC risks exposing the patient to a doublet they cannot benefit from.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639	no additional tissue if CGP is run; 1-2 unstained slides if VE1 IHC is added
IDH1 R132 hotspot detection by tumor DNA NGS (included in comprehensive panel)	Ivosidenib eligibility (ClarIDHy) requires a confirmed IDH1 R132 mutation, and the codon is captured by any comprehensive DNA panel, so this is satisfied by the shared comprehensive panel rather than a standalone assay. The row exists to make the IDH1 gate explicit and to note that IDH2 and non-R132 IDH1 variants do not qualify for ivosidenib. If only a limited fusion-focused panel is run, the IDH1 hotspot can be missed, so confirm DNA coverage when ordering.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639	no additional tissue if CGP is run

RNA-based fusion NGS for NTRK1/2/3 and RET (pan-TRK IHC as an optional screen)	NTRK and RET fusions are rare in biliary tract cancer (<1%) but tumor-agnostically actionable, and an RNA-based assay detects them more reliably than DNA-only calling, especially for partners with large intronic breakpoints. This is satisfied by the RNA arm of the shared paired panel; a standalone RNA fusion test or pan-TRK IHC screen is only needed if a DNA-only platform was used. A missed fusion forecloses a high-response oral targeted option.	Caris Life Sciences (Caris MI Profile (whole-transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 888-979-8669	no additional tissue if paired DNA/RNA CGP is run
KRAS G12C detection by tumor DNA NGS (included in comprehensive panel)	KRAS G12C is uncommon in biliary tract cancer but, when present, opens covalent G12C inhibitors and trial enrollment; the codon is covered by any comprehensive DNA panel, so the shared CGP resolves it. The row flags that broader KRAS hotspot results (e.g. G12D, G12V) also carry prognostic and emerging-trial relevance even though they do not gate a G12C inhibitor. No separate assay is needed if the ordered panel reports KRAS codon 12.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 888-988-3639	no additional tissue if CGP is run

Baseline and serial ctDNA for FGFR2 kinase-domain mutations and co-altered PIK3CA / CDKN2A/B	If FGFR2 fusion is confirmed and an FGFR inhibitor is started, acquired FGFR2 kinase-domain mutations (gatekeeper N549K, V564F and others) and co-alterations such as PIK3CA or CDKN2A/B drive resistance and can prompt a switch to a next-generation FGFR inhibitor. Serial ctDNA tracks these clones earlier than imaging or re-biopsy. This is sequencing intelligence rather than an enrollment gate, so it is not essential up front, but it should be in place before or shortly after starting therapy.	Guardant Health (Guardant360) · test info · 3100 Hanover Street, Palo Alto, CA 94304 · 855-698-8887	2 x 10 mL Streck tubes whole blood
Germline multigene panel (Lynch genes BRCA1/2; reflex from somatic findings)	If MMR IHC shows MLH1 loss or the somatic panel returns a pathogenic Lynch-gene or BRCA1/2 variant, germline testing distinguishes a heritable syndrome from a sporadic somatic event and triggers cascade family screening. MLH1 loss usually warrants a BRAF V600E or MLH1-promoter methylation check first to gauge whether germline testing is indicated. This does not change the patient's immediate therapy, so it is research-adjacent for the targeting question, but it carries real family implications when a driver is germline.	Invitae (Invitae Multi-Cancer / Lynch Syndrome Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 800-436-3037	5-10 mL whole blood or a saliva kit

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.